

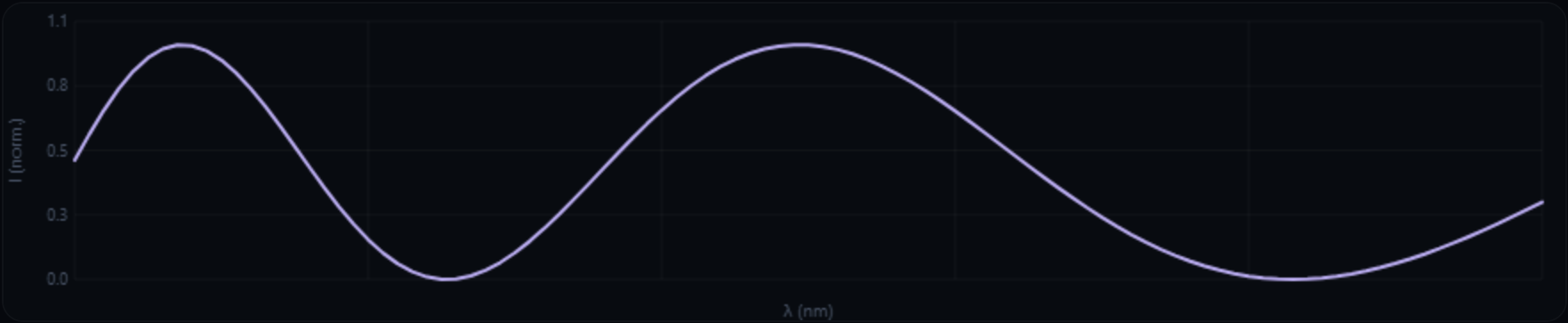
Thin-Film Interference

Wedge fringes: n, d, θ ; $\cos^2(\delta/2)$ colors and $I(\lambda)$ at mid thickness.

thinner ← $d(x)$ → thicker

Two-beam model: reflected rays from top and bottom of film interfere.

LIVE GRAPHS



FILM

Film index n 1.38

Min thickness d 80 nm

Max thickness d 1120 nm

ILLUMINATION

Angle in air θ 18 °

REFLECTION PHASES

Extra π : one common textbook case (e.g. asymmetric reflections) shifts bright/dark fringes.

Extra π off

Wedge along the strip: d grows left→right. Colors use $\cos^2(\delta/2)$ at three wavelengths ($\approx R, G, B$). Path in film uses Snell: $\cos \theta_f$ with $n \sin \theta_f = \sin \theta$.

MEASURED VALUES

$\cos \theta_f$

0.9746

About this model

Thin-film interference occurs when light waves reflecting from the top and bottom surfaces of a thin layer, such as a soap bubble or an oil slick, superpose. This simulator visualizes the interference pattern produced by a thin film of varying thickness, specifically a wedge-shaped film. The core physics principle is the phase difference, δ , accumulated between the two reflected rays. This phase difference depends on the optical path difference (OPD), which is a function of the film's thickness (d), its refractive index (n), the angle of incidence (θ), and the wavelength of light (λ). For near-normal incidence, the OPD is approximately $2nd$. A crucial detail is the π phase shift (or half-wavelength shift) that occurs when light reflects off a boundary from a lower-index to a higher-index medium; this simulator correctly accounts for this condition. The resulting intensity for a given wavelength is governed by $I = I_0 \cos^2(\delta/2)$, where $\delta = (2\pi/\lambda) * OPD + \pi$. The simulator shows the resulting colored fringes as the thickness varies, mapping the $\cos^2$ function to visible colors via an additive RGB model based on the spectral intensity distribution $I(\lambda)$. Key simplifications include assuming perfectly coherent, monochromatic light sources for the underlying calculation, ignoring effects of multiple reflections within the film (a valid approximation for low-reflectivity films), and modeling the wedge with a linear thickness gradient. By interacting with the controls for n , d , and θ , students learn to predict fringe spacing, understand how colors arise from constructive and destructive interference of different wavelengths, and directly see the relationship between optical path difference and observed intensity.

Who it's for: Undergraduate physics or engineering students studying wave optics, particularly the chapter on interference. It is also valuable for advanced high school students in AP Physics or IB courses.

KEY TERMS

- Thin-Film Interference
- Optical Path Difference
- Phase Shift
- Constructive Interference
- Destructive Interference
- Refractive Index
- Wedge Fringes
- Coherence

How it works

Light reflected from the top and bottom surfaces of a transparent film can interfere. The optical path difference is approximately $2 n d \cos \theta_f$ inside the film (θ_f from Snell's law). A wedge-shaped film produces rainbow-like fringes as d changes; turning the incidence angle or adding an extra π from reflection boundary conditions shifts the fringe pattern.

Key equations

$$n \sin \theta_f = \sin \theta \text{ (air} \rightarrow \text{film)}$$

$$\delta = 4\pi n d \cos \theta_f / \lambda + \delta_{\text{extra}} \cdot I \propto \cos^2(\delta/2)$$

Frequently asked questions

Why are the fringes straight and parallel in a wedge?

In an ideal wedge, the thickness (d) changes linearly along its length. Each fringe corresponds to a line of constant thickness where the optical path difference satisfies the condition for a maximum or minimum. Since lines of constant thickness are straight and parallel to the thin edge of the wedge, the resulting interference fringes are also straight and parallel.

Why do we see different colors instead of just bright and dark bands?

White light contains a continuous spectrum of wavelengths. For a given film thickness, some wavelengths interfere constructively (appear bright) while others interfere destructively (appear dim). Our eyes perceive the mixture of these enhanced and suppressed wavelengths as a specific color. This is why soap bubbles and oil slicks show vibrant colors, not just black and white.

What is the role of the ' π phase shift' mentioned in the model?

The π phase shift (equivalent to a half-wavelength path difference) occurs when light reflects off a boundary going from a medium with a lower refractive index to one with a higher index (e.g., from air to film). This extra shift flips the conditions for constructive and destructive interference. Without accounting for it, the predictions for bright and dark fringes would be incorrect. The simulator includes this shift automatically.

Does this simulator model all real-world thin films perfectly?

No, it makes several simplifying assumptions. It uses a two-beam interference approximation, which is excellent for films with low reflectivity (like soap films). It does not account for effects like absorption in the film, divergence of the light beam, or the precise polarization of light, which can influence real-world interference patterns.

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Michelson Interferometer

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Mach-Zehnder Interferometer

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Spherical lens on flat glass: air-film thickness fringes;

$I(\Delta) = V \cos^2(\pi\Delta/\lambda)$; tilt fringes; coherence length

Two-beam recombination; $I \propto \cos^2(\pi\Delta/\lambda)$ vs one-arm

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Anti-Reflection Coating

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Air–film–glass at normal incidence: $R(\lambda)$ from two-interface interference; quarter-wave design.

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Toy 2D interference $|E_{\text{ref}} + E_{\text{obj}}|^2$: fringes encode phase; readout diffraction not simulated.

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Random phased waves: $|\sum e^{i(k\cdot r + \phi)}|^2$ grain pattern (qualitative).